

Ahmet Taha Albayrak

AI ENGINEER

Ankara, Türkiye

ahmetahalbayrak@gmail.com | LinkedIn | GitHub | Portfolio

PROFESSIONAL SUMMARY

AI Engineer with 8+ years of experience designing, optimizing, and deploying production machine-learning systems across computer vision, multimodal generative AI, agentic workflows, edge inference, and large-scale image processing. Experienced in model architecture, performance optimization, system integration, and technical delivery.

SELECTED IMPACT

+30% / -40%

Accuracy / compute

Multi-task perception

30 FPS / 3x

Edge inference

Real-time optimization

1B+

Image records

Large-scale ML pipelines

EXPERIENCE

NuveMind Information Technologies

Mar 2025 - Present

Ankara, Türkiye

AI ENGINEER

Designing and integrating multimodal generative AI and agentic systems for production use.

- Architect multimodal pipelines integrating image-processing, language-model, and product-service components.
- Define tool interfaces and orchestration patterns for agents executing multi-step API workflows.
- Coordinate implementation across AI and software teams, defining integration and delivery practices.

Pavo Group

Nov 2021 - Jan 2025

Ankara, Türkiye

SENIOR MACHINE LEARNING ENGINEER

Developed and optimized production machine-learning systems for perception and edge deployment.

- Developed multi-task architectures for 2D/3D pose estimation and object detection, improving accuracy by 30% while reducing computational cost by 40%.
- Optimized real-time edge inference to 30 FPS, delivering a threefold performance improvement.
- Integrated IR-camera data into ML pipelines, increasing low-light data-collection capacity by 50%.

TÜBİTAK Space Research Institute

Jul 2020 - Nov 2021

Ankara, Türkiye

RESEARCHER

Conducted machine-learning research in distributed computing, containerized services, and algorithm development.

- Investigated volunteer computing for hyperparameter optimization and published the results at the 6th High Performance Computing Conference.
- Containerized microservices with Docker to improve deployment consistency and scalability.
- Developed machine-learning algorithms using Python, PyTorch, and MXNet.

GCLab Engineering

Oct 2018 - Jul 2020

Ankara, Türkiye

COMPUTER ENGINEER

Developed computer-vision and multimodal retrieval systems with the TÜBİTAK UZAY image-processing team.

- Developed vision-based systems and image-processing algorithms for research applications.
- Implemented faster and more compact semantic indexing for image and audio data.
- Supported multimodal user queries in internet-facing search systems.

SELECTED PROJECTS

1bitllm

2026

Browser-native playground for quantized Bonsai language models using ONNX inference, WebGPU, token streaming through a Web Worker, and local model caching without a server-side inference backend.

Engineering contribution: Designed and implemented the client-side inference flow, model lifecycle, streaming chat interface, and WebGPU capability checks.

WebGPU / Transformers.js / ONNX / Web Workers

Live projectSource

OpenTTD / Browser Port

2019

Browser port of OpenTTD compiled with Emscripten and WebAssembly, bringing the C++ simulation engine to a full-screen web canvas without a native installation.

Engineering contribution: Ported and packaged OpenTTD for browser execution, integrating the WebAssembly runtime, asset bundle, loading states, and canvas-based application shell.

C++ / WebAssembly / Emscripten / Canvas

Live projectSource

EDUCATION

Middle East Technical University

Sep 2021 - Sep 2023

Graduate studies in Computer Engineering

CGPA 3.34 / 4.00 - on leave since Sep 2023

Bilkent University

Sep 2013 - Jun 2018

B.Sc. in Computer Engineering

CGPA 3.32 / 4.00

PUBLICATION

A.T. Albayrak, I. Atıl, "Hyper Parameter Optimization for Deep Learning using Volunteer Computing," 6th High Performance Computing Conference, 2020.

RECOGNITION

- Bilkent University Senior Project Data Science Award (2017-2018)
- Ranked 753rd in the quantitative track among approximately 1.8 million candidates in Türkiye's national university entrance examination
- Enderun Science High School Salutatorian

TECHNICAL SKILLS

PROGRAMMING	Python, C++, C, MATLAB, SQL
MACHINE LEARNING	PyTorch, TensorFlow, MXNet, NumPy, CUDA
INFRASTRUCTURE	Docker, Git, MLOps, Microservices
SPECIALIZATION	Computer Vision, Multimodal AI, Agentic AI, Edge AI, Semantic Search
LANGUAGES	Turkish - Native; English - Professional working proficiency